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Unit 1 - Exploring the Core

PDF Documents

PDF Document 1: 1738066994726-Chapter 5_unit 1 - Exploring the Core.pdf

This PDF document is included in the unit materials.



UNIT 1 EXPLORING THE CORE

Topic Investigation

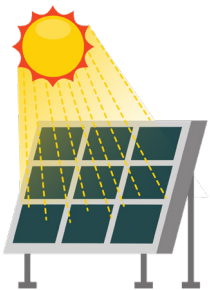


EXPLORING RENEWABLE ENERGY

Welcome to the first unit of our journey into renewable energy, a key component in fighting climate change and creating a sustainable future. In this unit, we'll discover the different types of renewable energy sources and understand why they are essential for our planet.

TYPES OF RENEWABLE ENERGY

Renewable energy comes from natural sources that continuously replenish themselves. The main types we'll cover include:



SOLAR POWER

Energy from the sun. This is harnessed using solar panels that convert sunlight directly into electricity.

WIND ENERGY

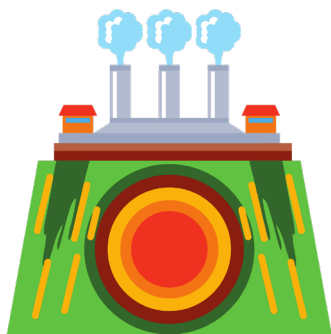
Energy generated by the wind. Wind turbines capture kinetic energy from the wind and convert it into electricity.





HYDROELECTRIC POWER

Energy from water. This type of energy uses water flow, typically from rivers or dams, to generate power.

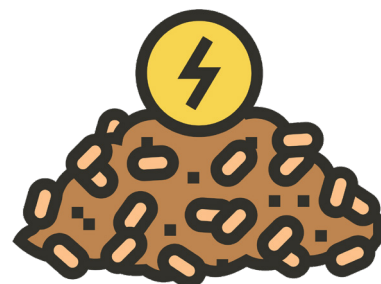


GEO THERMAL ENERGY

Energy from the heat beneath the earth's surface. This can be harnessed to produce heat or electricity.

BIOMASS ENERGY

Energy from organic materials. Plants, agricultural crops, and waste materials can be used to create bioenergy.



WHY RENEWABLE ENERGY?

Renewable energy is crucial for reducing our carbon footprint. Unlike fossil fuels, renewable sources produce little to no greenhouse gases or pollutants. Here's why embracing renewable energy is essential:

- **Environmental Impact:** Reduces the effects of global warming and air pollution.
- **Sustainability:** Ensures a long-lasting energy supply without depleting natural resources.
- **Economic Benefits:** Creates jobs in new and emerging green technologies.

THE SCIENCE BEHIND RENEWABLE ENERGY

Each type of renewable energy relies on specific scientific principles to convert natural phenomena into usable power. For example:

- **SOLAR PANELS** use photovoltaic cells to convert sunlight into electricity.
- **WIND TURBINES** use aerodynamic forces to turn rotor blades that drive generators to produce electricity.



Inquiry & Reflection

Discussion question 1

Which type of renewable energy do you think is most effective for our local community? Why?

Discussion question 2

How could increasing the use of renewable energy change our daily lives and local environment?

